Measuring Lexical Reachability Across Generator Tiers via the Word Coverage Score

Anonymous ACL submission

Abstract

Standard evaluations of large language models (LLMs) focus on task accuracy or aggregate perplexity, leaving a structural question underexplored: when a model generates or evaluates text, how much of a reference response's vocabulary remains reachable inside the probability-mass truncation filters used during decoding? We operationalize *lexical reachability* using the **Word Coverage Score (WCS)** (Anonymous, 2026), the fraction of words in a response whose constituent sub-tokens all survive a scorer's nucleus or top-k candidate-pool filter under teacher forcing. Auditing a cross-model matrix on HC3 ELI5 ($N = 50$ prompts; 4 core `bf16` open-weight scorers and `Qwen3.5-27B-int8` across 16 human, frontier proprietary, large open-weight, and small open-weight sources) and replicating across a four-family expanded pool (10 scorers across Qwen, Gemma, Llama, and Mistral up to `Gemma-4-31B-it` in `bf16`; 21 sources) as well as HC3 Wikipedia CS/AI ($N = 50$), we uncover four consistent empirical patterns. First, reachability follows a strict four-tier ordering under standard top-K nucleus filtering ($p = 0.99, K = 100$): **Human** (0.8374) $<$ **Frontier Proprietary** (0.8627; $\Delta = +0.0253$, 95% CI $[+0.0163, +0.0346]$) $<$ **Large Open-Weight** (0.8862; $\Delta = +0.0235$ over Frontier Proprietary, 95% CI $[+0.0210, +0.0261]$) $<$ **Small Open-Weight** (0.9602; $\Delta = -0.0740$ for Large minus Small Open-Weight, 95% CI $[-0.0772, -0.0708]$). Second, a candidate-support expansion audit at $K = 1000$ and exact full-vocabulary nucleus scoring (`strict_full`) reveals that the gap between Human and Frontier Proprietary text is an artifact of top-K rank truncation on human long-tail lexical choices: under `strict_full` ($p = 0.99$), Human reachability rises to 0.8784, matching Frontier Proprietary (0.8730; $\Delta = +0.0054$, 95% CI $[-0.0025, +0.0131]$), whereas the separation between Frontier/Large Open-Weight and Small Open-Weight models remains large ($\Delta = -0.0687$, 95% CI $[-0.0713, -0.0662]$) across every filter regime. Third, excluding self-scoring cells ($s \neq g$), open-weight scorers exhibit a persistent **Same-Family Advantage** ($\Delta = +0.0316$, 95% CI $[+0.0294, +0.0343]$ across Qwen/Gemma pairs; $+0.0234$, 95% CI $[+0.0214, +0.0257]$ across all four families) that widens under tighter nucleus thresholds ($p = 0.95, 0.90$) and remains $> 98\%$ intact after excluding family-specific orthographic confounds. Fourth, a Zipf-adjusted lexical loss taxonomy across 2.59M word evaluations demonstrates that open-weight models achieve high mutual reachability by pruning low-probability content words and discourse markers rather than function words.

1 Introduction

Modern large language models (LLMs) rarely sample from their full softmax distribution over vocabulary V. In both interactive deployments and synthetic data pipelines, generation is governed by candidate-pool truncation operators—such as top-k filtering, nucleus (top-p) sampling (Holtzman et al., 2020), locally typical sampling (Meister et al., 2023), and temperature scaling—that zero out the probability tail at every decoding step. While truncation prevents degenerate tail excursions, it also imposes a hard support boundary: any word whose constituent sub-tokens fall outside the active candidate pool at even a single token position becomes *unreachable* under that decoding filter.

Understanding which words survive or vanish across model families and scale tiers matters for three practical settings: (i) **synthetic data and distillation**, where student models trained on teacher outputs inherit the lexical support of the teacher's decoding filter (Jiang et al., 2025; Savani et al., 2025); (ii) **speculative decoding**, where acceptance rates depend on whether target-model tokens reside within the draft model's high-probability support (Leviathan et al., 2023; Chen

et al., 2023); and (iii) **machine-generated text forensics**, where zero-shot detectors exploit token-rank and probability-curvature concentration relative to human writing (Gehrmann et al., 2019; Mitchell et al., 2023; Su et al., 2023; Bao et al., 2024; Hans et al., 2024).

Despite extensive work on perplexity and scalar log-likelihood ratios, sequence-level averages obscure *support survival*: a response can have moderate average log-probability while containing specific content words that a reference model's truncation filter would deterministically reject. To measure word-level support survival directly, we build on the **Word Coverage Score (WCS)** (Anonymous, 2026), defined as the proportion of whitespace-delimited words in a source text whose overlapping sub-tokens all survive a scorer model's candidate-pool filter under teacher-forced conditioning on the ground-truth prefix (Figure 1).

We conduct a systematic cross-model WCS matrix audit across two benchmark domains from the Human ChatGPT Comparison Corpus (HC3; Guo et al., 2023)—open-ended explanatory answers (**ELI5**, $N = 50$) and technical encyclopedia definitions (**Wikipedia CS/AI**, $N = 50$)—spanning 16 primary generators (1 human, 3 frontier proprietary, 7 large open-weight, and 5 small open-weight) and extending to a 21-source, 10-scorer four-family suite covering Qwen 3.5/2.5, Gemma 4 (up to 31B in `bf16`), Llama 3.1/3.2, and Mistral v0.3. Our contributions are fourfold:

1. **Four-tier lexical reachability hierarchy:** Under standard realistic nucleus filtering ($p = 0.99, K = 100$), WCS stratifies generators into four ordered tiers: Human (0.8374) $<$ Frontier Proprietary (0.8627) $<$ Large Open-Weight (0.8862) $<$ Small Open-Weight (0.9602), with statistically significant prompt-paired bootstrap separations between every adjacent tier (Table 3).

2. **Resolution of the Human vs. Frontier gap via support expansion:** By comparing $K = 100$, $K = 1000$, and exact full-vocabulary nucleus filtering (`strict_full`), we prove that the lower WCS of human text relative to Frontier Proprietary models at $K = 100$ is driven by top-100 rank misses on rare human word choices rather than cumulative probability mass exceeding $p = 0.99$. Under `strict_full` ($p = 0.99$), Human (0.8784) and Frontier Proprietary (0.8730) reachability converge ($\Delta = +0.0054$, 95% CI $[-0.0025, +0.0131]$), whereas Small Open-Weight models remain $+0.0687$ higher than Large Open-Weight models and $+0.0833$ higher than Human writing.

3. **Cross-family generalization and Same-Family Advantage:** Expanding the audit to 10 scorers and 21 sources across four model families (Qwen, Gemma, Llama, Mistral) and replicating on Wikipedia CS/AI confirms that both the tier hierarchy and an off-diagonal **Same-Family Advantage** ($+0.0234$ to $+0.0316$ at $p = 0.99$, rising to $+0.0526$–$+0.0580$ at $p = 0.90$) hold across architectures and domains, and are invariant ($> 98\%$ preserved) to excluding family-specific orthographic tokens.

4. **Zipf-adjusted lexical loss taxonomy:** Auditing 2.59M off-diagonal word evaluations reveals that while function words ('the', 'a', 'and') top raw lost-word counts solely due to Zipfian base frequency (94%–98% survival), per-occurrence loss rates are $2.1\times$–$2.9\times$ higher on content/lexical words across every tier.

2 Related Work

Truncation Sampling and Lexical Diversity. Holtzman et al. (2020) introduced nucleus (top-p) sampling to dynamically bound the active vocabulary at each decoding step, mitigating the incoherent tail selections of pure ancestral sampling while avoiding the repetitive loops of greedy or beam search. Subsequent work proposed locally typical sampling (Meister et al., 2023) and adaptive truncation schedules, while production LLM releases ship model-specific combinations of temperature T, top-k, and top-p in their generation configurations (Qwen Team, 2026; Google and Hugging Face, 2026). Rather than proposing a new decoding heuristic, we audit how existing truncation filters act as a cross-model lexical sieve.

Zero-Shot Detection and Cross-Model Scoring. Statistical detectors of machine-generated text—including GLTR (Gehrmann et al., 2019), DetectGPT (Mitchell et al., 2023), DetectLLM (Su et al., 2023), Fast-DetectGPT (Bao et al., 2024), DNA-GPT (Yang et al., 2024), and Binoculars (Hans et al., 2024)—rely on the observation that

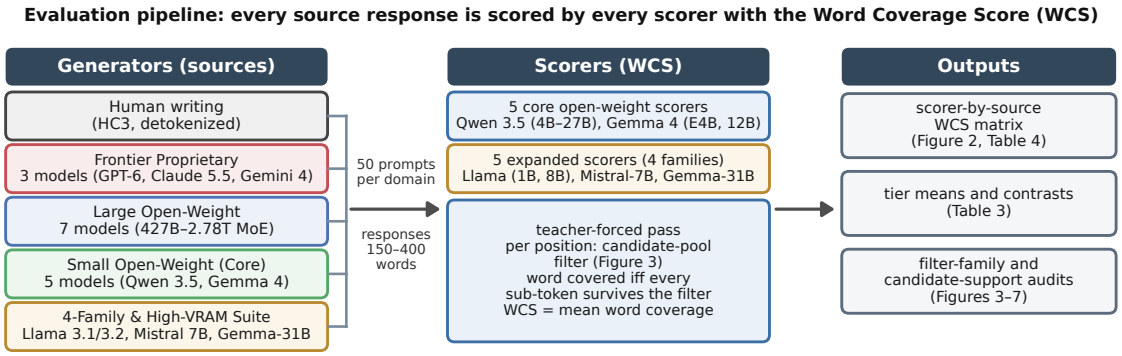


Figure 1: Overview of the cross-model Word Coverage Score (WCS) evaluation pipeline. Responses from four generator tiers—Human reference answers (detokenized HC3), Frontier Proprietary APIs (3 models), Large Open-Weight APIs (7 models, 427B–2.78T), and Small Open-Weight local models (core Qwen 3.5 and Gemma 4 suite plus 4-family Llama 3 and Mistral expansion)—are evaluated under teacher forcing by open-weight scorers across multiple candidate-pool filter families.

machine-generated tokens concentrate in the high-probability head of a scorer's predictive distribution. Recent benchmark studies (Dugan et al., 2024; Koike et al., 2025) show that detector sensitivity varies substantially with generator scale, decoding temperature, and scorer-generator family alignment. Our cross-model WCS matrix complements detection benchmarks by decomposing token-level survival into interpretable word-level reachability across four distinct generator tiers and multiple filter regimes.

Corpus Homogeneity and Distillation Signatures. Jiang et al. (2025) documented the "Artificial Hivemind" phenomenon, showing that open-ended generations across diverse LLMs converge toward homogeneous lexical and semantic patterns. Relatedly, Savani et al. (2025) studied how sampling modifications alter the distillability of teacher distributions. Our results provide a direct distributional mechanism for these observations: smaller open-weight models confine their generations to a shared, high-probability core vocabulary that survives even aggressive cross-family truncation filters (83%–96% WCS), whereas human writers and frontier proprietary models draw substantially deeper from the lexical tail.

3 Methodology: Word Coverage Score

Token-Level Nucleus Survival. Let a scorer language model g with vocabulary V_g and fast subword tokenizer $\mathcal{T}_g$ evaluate a response text y conditioned on a chat-formatted prompt prefix x. Concatenating x and y yields token sequence $(t_1, \ldots, t_m)$, where answer token positions $I_y \subset \{1, \ldots, m\}$ are identified via character offset mappings so that every token overlapping the response body is scored conditioned on its exact ground-truth prefix $t_{<i}$.

At each answer position $i \in I_y$, the scorer produces logits $z_i \in \mathbb{R}^{|V_g|}$ and temperature-scaled probabilities $P_i(v) = \text{softmax}(z_i/T)_v$. Let π_i be the permutation sorting V_g in descending order of P_i, with sorted probabilities $p_{i,(1)} \geq p_{i,(2)} \geq \cdots \geq p_{i,(|V_g|)}$. We evaluate four candidate-pool filter families $\mathcal{F}$ (illustrated in Figure 3):

1. **Realistic top-p over top-K (`realistic`):** In stored-logit audits and standard two-stage samplers, the candidate support is first restricted to the top-K tokens ($K = 100$ primarily, $K = 1000$ in validation) and renormalized:

$$\tilde{p}_{i,(r)} = \frac{p_{i,(r)}}{\sum_{j=1}^{K} p_{i,(j)}}, \quad 1 \leq r \leq K. \qquad (1)$$

   The observed token $t_i = \pi_i(r_i)$ survives at threshold $p \in (0, 1]$ iff $r_i \leq K$ and the strictly preceding renormalized cumulative mass satisfies $\sum_{j=1}^{r_i-1} \tilde{p}_{i,(j)} \leq p$ (which retains the boundary token crossing p).

2. **Strict unnormalized top-K (`strict`):** Retains t_i iff $r_i \leq K$ and the unnormalized preceding cumulative mass satisfies $\sum_{j=1}^{r_i-1} p_{i,(j)} \leq p$.

3. **Exact full-vocabulary nucleus (`strict_full`):** Removes the top-K

rank cap entirely ($K = |V_g|$) and retains t_i iff the exact full-vocabulary preceding cumulative mass satisfies

$$M_i(t_i) = \sum_{v \in V_g : P_i(v) > P_i(t_i)} P_i(v) \leq p. \quad (2)$$

Because $M_i(t_i)$ is computed online on the GPU from the full softmax tensor before storing top-K arrays, `strict_full` is exact with zero truncation error.

4. **Shipped generation defaults (`default`):** Applies each scorer's released generation hyperparameters (T_g, k_g, p_g), scaling top-k_g probabilities by $p_{i,(r)}^{1/T_g}$, renormalizing over k_g, and retaining tokens whose preceding cumulative mass is $\leq p_g$.

Word-Level Aggregation. Because different model families employ distinct BPE tokenizers ($|V_g|$ ranging from 32,768 in Mistral-7B to 262,144 in Gemma 4), token-level survival rates are confounded by subword fertility: a rare compound word split into four sub-tokens has four opportunities to fail. Following Anonymous (2026), we partition the response y into whitespace-delimited words $W(y) = (w_1, \ldots, w_L)$ via Unicode regex `\S+`. Each word w_ℓ with character span $[a_\ell, b_\ell)$ maps to the non-empty set of overlapping answer token positions $G_\ell = \{i \in I_y : [c_i, d_i) \cap [a_\ell, b_\ell) \neq \emptyset\}$. Word w_ℓ survives filter $\mathcal{F}$ iff *every* constituent sub-token survives:

$$\text{surv}(w_\ell; g, \mathcal{F}) = \prod_{i \in G_\ell} \mathbb{I}\big[t_i \in \mathcal{F}(P_i)\big]. \quad (3)$$

The prompt-level **Word Coverage Score** is the fraction of surviving words in response y (ranging in $[0, 1]$):

$$\text{WCS}(y; g, \mathcal{F}) = \frac{1}{|W(y)|} \sum_{\ell=1}^{|W(y)|} \text{surv}(w_\ell; g, \mathcal{F}). \quad (4)$$

Self-Scoring Invariant. When source s generates a response via greedy decoding ($\arg\max_v P_i(v)$ at every step) and is audited by the identical model $g = s$ under matched chat formatting and precision, every token has rank $r_i = 1$ and preceding probability mass $0 \leq p$, guaranteeing $\text{WCS}(y; s, \mathcal{F}) = 1.0000$. Any deviation from 1.00 on the diagonal therefore diagnoses either BPE re-tokenization merges across multi-token strings or numerical tie-breaking in quantized kernels.

Paired Bootstrap Inference. Because word coverage varies substantially across prompt topics, all source and tier comparisons are paired at the prompt level. For two sources or source tiers A and B evaluated across N shared prompts, let $\bar{u}_q^A$ and $\bar{u}_q^B$ denote the mean WCS on prompt q (averaged across the active scorer pool and tier sources). The point contrast is $\hat{\Delta} = \frac{1}{N} \sum_{q=1}^{N} (\bar{u}_q^A - \bar{u}_q^B)$, and we report two-sided 95% confidence intervals from $B = 10{,}000$ paired bootstrap resamples over the N prompts (fixed seed 42).

4 Experimental Setup

Scorer and Generator Pools. Tables 1 and 2 summarize our model pools. To ensure that primary scorer averages are never confounded by low-bit weight quantization, our primary core scorer pool comprises the four unquantized `bf16` checkpoints from the Qwen 3.5 and Gemma 4 families (`qwen-4b`, `qwen-9b`, `gemma-e4b`, `gemma-12b`), while `qwen-27b-int8` (27B in 8-bit LLM.int8) is audited in parallel and reported in all matrix tables. Our smallest evaluated Qwen and Gemma checkpoints begin at 4B and E4B because preliminary 2B/E2B checkpoints violated the 150–400 word length window on a substantial fraction of prompts and were excluded from the evaluation pool. To test architectural generality beyond Qwen and Gemma, our four-family expansion adds `Qwen2.5-32B-Instruct` (`int8`), `Llama-3.2-1B-Instruct` (`bf16`), `Llama-3.1-8B-Instruct` (`bf16`), and `Mistral-7B-Instruct-v0.3` (`bf16`), yielding 9 scorers (7 `bf16` + 2 `int8`) and 20 sources. Finally, to test unquantized `bf16` scaling at the $\sim$ 30B tier without low-bit weight quantization, we evaluate `Gemma-4-31B-it` (31.3B, requiring $\sim$ 58 GiB in `bf16`) on a dual-GPU 96 GiB high-VRAM server, bringing the complete evaluation suite to 10 scorers (8 `bf16` + 2 `int8`) and 21 sources across HC3 ELI5 and Wikipedia CS/AI.

Benchmark Domains and Detokenization Normalization. We sample $N = 50$ frozen prompts (seed 42) from two subsets of HC3 (Guo et al., 2023) whose reference human answers contain 150–400 words: **HC3 ELI5** (explanatory lay-science questions) and **HC3 Wikipedia CS/AI** (encyclopedic computer science and artificial intelligence entries). Every model is prompted with the identical instruction: *"Answer the following question*

Model Name	Family	HuggingFace Repository ID	Pinned Revision	Params	Precision	Vocab $\lvert V \rvert$	Shipped Default (T, k, p)
Core Scorer & Local Generator Pool (HC3 ELI5 & Wikipedia CS/AI)							
`qwen-4b`	Qwen 3.5	`Qwen/Qwen3.5-4B`	`851bf6e8`	4.0B	`bf16`	248,320	$T = 0.6, k = 20, p = 0.95$
`qwen-9b`	Qwen 3.5	`Qwen/Qwen3.5-9B`	`c2022362`	9.0B	`bf16`	248,320	$T = 0.6, k = 20, p = 0.95$
`qwen-27b-int8`†	Qwen 3.5	`Qwen/Qwen3.5-27B`	`fc05daec`	27.0B	`int8`	248,320	$T = 0.6, k = 20, p = 0.95$
`gemma-e4b`	Gemma 4	`google/gemma-4-E4B-it`	`ee0ef602`	8.0B	`bf16`	262,144	$T = 1.0, k = 64, p = 0.95$
`gemma-12b`	Gemma 4	`google/gemma-4-12B-it`	`707f0a3b`	12.0B	`bf16`	262,144	$T = 1.0, k = 64, p = 0.95$
Four-Family Cross-Architecture Expansion Pool							
`qwen-32b-int8`†	Qwen 2.5	`Qwen/Qwen2.5-32B-Instruct`	`5ede1c97`	32.5B	`int8`	152,064	$T = 0.7, k = 20, p = 0.80$
`llama-3.2-1b`	Llama 3.2	`meta-llama/Llama-3.2-1B-Instruct`	`92131767`	1.2B	`bf16`	128,256	$T = 0.6, k = 100, p = 0.90$
`llama-3.1-8b`	Llama 3.1	`meta-llama/Llama-3.1-8B-Instruct`	`0e9e39f2`	8.0B	`bf16`	128,256	$T = 0.6, k = 100, p = 0.90$
`mistral-7b`	Mistral 0.3	`mistralai/Mistral-7B-Instruct-v0.3`	`c170c708`	7.2B	`bf16`	32,768	$T = 1.0, k = 100, p = 0.95$
High-VRAM Server Extension (> 60 GiB GPU Required for Unquantized `bf16`)							
`gemma-31b`	Gemma 4	`google/gemma-4-31B-it`	`842da379`	31.3B	`bf16`	262,144	$T = 1.0, k = 64, p = 0.95$

Table 1: Open-weight local models serving as both teacher-forced WCS scorers and local greedy/sampled generators. Primary tier means use the 4 unquantized `bf16` core scorers (`qwen-4b`, `qwen-9b`, `gemma-e4b`, `gemma-12b`), while `qwen-27b-int8`† and `qwen-32b-int8`† are reported alongside to isolate 8-bit quantization effects. Sub-4B Qwen/Gemma checkpoints (`Qwen3.5-2B`, `Gemma-4-E2B`) failed preliminary length-adherence checks on the 150–400 word constraint and were excluded prior to matrix aggregation.

Generator Name	Provider / Family	Total / Active Params	Tier
`gpt-6-astra`	OpenAI	Proprietary	Frontier Prop.
`claude-opus-5.5`	Anthropic	Proprietary	Frontier Prop.
`gemini-4-argon`	Google	Proprietary	Frontier Prop.
`qwen3.8-2.4t-a95b`	Qwen 3.8 MoE	2.4T / 95B	Large Open-Wt.
`deepseek-v4-pro-0813`	DeepSeek V4 Pro	1.68T / 64B	Large Open-Wt.
`deepseek-v4.1-flash-1017`	DeepSeek V4.1 Flash	890B / 42B	Large Open-Wt.
`glm-5.3`	Z.ai GLM	1.15T / 58B	Large Open-Wt.
`kimi-k3`	Moonshot Kimi	2.78T / 110B	Large Open-Wt.
`minimax-m3`	MiniMax	427B / 45B	Large Open-Wt.
`mimo-v2.6-pro`	Xiaomi MiMo	960B / 48B	Large Open-Wt.

Table 2: Frontier Proprietary (3) and Large Open-Weight (7) source generators queried via OpenRouter API ($T = 1.0, p = 0.95$, reasoning traces excluded; $50/50$ complete responses on both HC3 ELI5 and Wikipedia CS/AI).

in English, using 150–400 words. Give only your answer." Crucially, raw HC3 ELI5 human answers contain legacy Penn Treebank / Moses whitespace artifacts before punctuation (e.g., "`word ,`" or "`don 't`") that artificially inflate whitespace word counts and create spurious standalone punctuation words. We apply standard Moses detokenization (`human-detok.jsonl`) before all ELI5 human audits so that human and model responses share identical orthographic conventions.

5 Results

5.1 Generator Tier Hierarchy on HC3 ELI5

Figure 2 and Tables 3–4 present our primary results on HC3 ELI5 under realistic nucleus filtering ($p = 0.99, K = 100$). Across the 4 core `bf16` scorers, mean WCS follows a monotonic four-tier hierarchy:

- **Human writing (detokenized HC3):** 0.8374 (16.26% of words lost outside the $p = 0.99$ top-100 nucleus).
- **Frontier Proprietary (3 models):** 0.8627 ($\Delta = -0.0253$ for Human minus Frontier Proprietary, 95% CI $[-0.0346, -0.0163]$), spanning GPT-6 Astra (0.8255, the only model source with lower reachability than human text, $\Delta = +0.0119$, 95% CI $[+0.0018, +0.0218]$), Claude Opus 5.5 (0.8728), and Gemini 4 Argon (0.8897).
- **Large Open-Weight (7 models, 427B–2.78T):** 0.8862, exceeding Frontier Proprietary by $+0.0235$ (95% CI $[+0.0210, +0.0261]$) and Human writing by $+0.0488$ (95% CI $[+0.0395, +0.0581]$). Every individual Large Open-Weight generator—from DeepSeek-V4.1-Flash (0.8648) to Qwen3.8-2.4T-A95B (0.9098)—lies above the Frontier Proprietary mean and far below every Small Open-Weight source.
- **Small Open-Weight (5 local models, 4B–27B):** 0.9602, exceeding Large Open-Weight by $+0.0740$ (Large minus Small $\Delta = -0.0740$, 95% CI $[-0.0772, -0.0708]$) and Human writing by $+0.1228$ (95% CI $[+0.1133, +0.1326]$). Even excluding self-scoring diagonal cells ($s \neq g$), Small Open-Weight reachability remains 0.9496.

Truncating every response to its first 150 whitespace words (`prefix150`, Table 3) leaves all four tier means $(0.8267, 0.8556, 0.8776, 0.9579)$ and pairwise contrasts ($\Delta = -0.0804$ Large vs. Small OW; $+0.0219$ Large vs. Frontier) virtually unchanged, ruling out response-length confounding.

5.2 Filter Sensitivity and Shipped Defaults

Figures 4–5 and Tables 5–6 (together with Tables 11–12 in the Appendix) trace how the tier

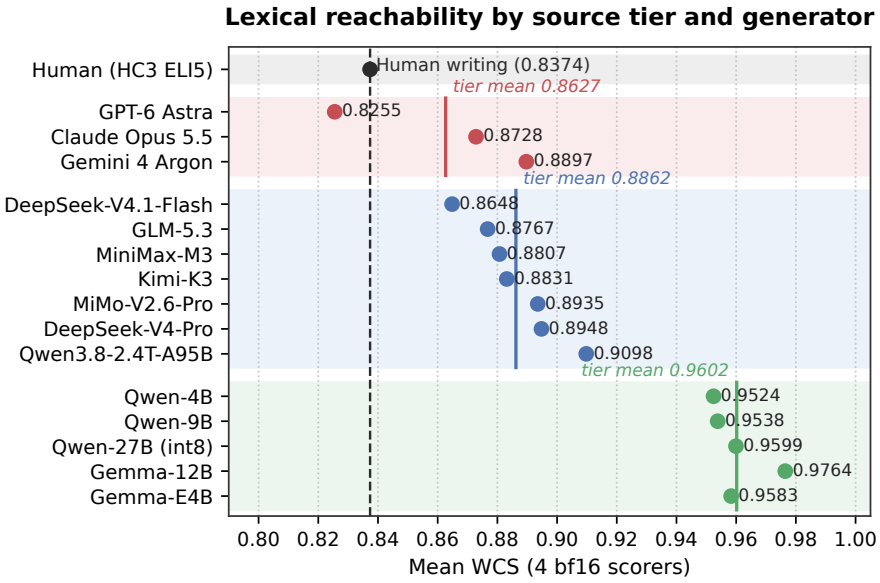


Figure 2: Mean Word Coverage Score (WCS) across the 4 core `bf16` scorers on HC3 ELI5 under the primary realistic filter ($p = 0.99, K = 100$). Generators stratify cleanly into four tiers: Human (0.8374) < Frontier Proprietary (0.8627) < Large Open-Weight (0.8862) < Small Open-Weight (0.9602).

Evaluation Suite / Filter Regime	Human	Frontier Prop.	Large OW	Small OW	Large − Small OW Δ (95% CI)	Large − Frontier Δ (95% CI)	Human − Frontier Δ (95% CI)
Primary Benchmark: HC3 ELI5 (4 Core bf16 Scorers, 16 Sources)							
Realistic ($p = 0.99$, primary)	0.8374	0.8627	0.8862	0.9602	−0.0740 ([−0.0771, −0.0708])	+0.0235 ([+0.0210, +0.0262])	−0.0253 ([−0.0346, −0.0163])
Realistic ($p = 0.95$)	0.7539	0.7761	0.8057	0.9170	−0.1113 ([−0.1158, −0.1068])	+0.0297 ([+0.0267, +0.0329])	−0.0222 ([−0.0334, −0.0113])
Realistic ($p = 0.90$)	0.6963	0.7217	0.7528	0.8856	−0.1328 ([−0.1382, −0.1274])	+0.0311 ([+0.0276, +0.0348])	−0.0255 ([−0.0375, −0.0136])
Strict unnormalized ($p = 0.99$)	0.8519	0.8689	0.8902	0.9611	−0.0708 ([−0.0737, −0.0680])	+0.0213 ([+0.0188, +0.0238])	−0.0171 ([−0.0258, −0.0086])
Full-vocabulary `strict_full` ($p = 0.99$)	0.8784	0.8730	0.8930	0.9617	−0.0687 ([−0.0713, −0.0661])	+0.0200 ([+0.0176, +0.0224])	+0.0054 ([−0.0025, +0.0131])
Shipped generation defaults	0.6591	0.7012	0.7359	0.8831	−0.1472 ([−0.1534, −0.1409])	+0.0348 ([+0.0308, +0.0389])	−0.0421 ([−0.0560, −0.0286])
Model-card non-thinking defaults	0.6144	0.6611	0.6947	0.8546	−0.1600 ([−0.1665, −0.1534])	+0.0336 ([+0.0292, +0.0381])	−0.0467 ([−0.0611, −0.0327])
Unscaled defaults ($T = 1.0$)	0.7157	0.7608	0.7934	0.9130	−0.1195 ([−0.1248, −0.1142])	+0.0326 ([+0.0293, +0.0362])	−0.0451 ([−0.0579, −0.0326])
150-word prefix truncation ($p = 0.99$)	0.8267	0.8556	0.8776	0.9579	−0.0803 ([−0.0841, −0.0765])	+0.0220 ([+0.0185, +0.0255])	−0.0289 ([−0.0379, −0.0201])
Cross-Family, Cross-Domain, and Decoding Regime Extensions							
HC3 ELI5 Sampled Local ($T = 1.0, p = 0.95$, 4 bf16 scorers)	0.8374	0.8627	0.8862	0.9513	−0.0651 ([−0.0686, −0.0617])	+0.0235 ([+0.0210, +0.0262])	−0.0253 ([−0.0346, −0.0163])
HC3 ELI5 4-Family Suite (9 scorers, 20 sources, $p = 0.99$)	0.8669	0.8952	0.9124	0.9599	−0.0475 ([−0.0503, −0.0446])	+0.0172 ([+0.0152, +0.0193])	−0.0283 ([−0.0357, −0.0209])
HC3 ELI5 4-Family Suite (7 bf16 scorers, 20 sources, $p = 0.99$)	0.8545	0.8847	0.9021	0.9555	−0.0535 ([−0.0565, −0.0504])	+0.0174 ([+0.0151, +0.0196])	−0.0303 ([−0.0384, −0.0223])
HC3 ELI5 4-Family Suite (9 scorers, $p = 0.95$)	0.7902	0.8128	0.8352	0.9102	−0.0750 ([−0.0790, −0.0710])	+0.0224 ([+0.0199, +0.0249])	−0.0226 ([−0.0320, −0.0135])
HC3 ELI5 4-Family Suite (9 scorers, `strict_full` $p = 0.99$)	0.9095	0.9104	0.9232	0.9627	−0.0395 ([−0.0415, −0.0373])	+0.0128 ([+0.0112, +0.0145])	−0.0009 ([−0.0061, +0.0042])
HC3 Wikipedia CS/AI (4 bf16 scorers, 15 sources, $p = 0.99$)	0.8571	0.8833	0.9042	0.9626	−0.0584 ([−0.0616, −0.0553])	+0.0209 ([+0.0184, +0.0235])	−0.0262 ([−0.0321, −0.0202])
HC3 Wikipedia CS/AI (4 bf16 scorers, $p = 0.95$)	0.7770	0.8116	0.8355	0.9245	−0.0890 ([−0.0937, −0.0842])	+0.0238 ([+0.0206, +0.0271])	−0.0346 ([−0.0420, −0.0271])
HC3 Wikipedia CS/AI (4 bf16 scorers, `strict_full` $p = 0.99$)	0.8872	0.8897	0.9088	0.9639	−0.0551 ([−0.0578, −0.0524])	+0.0191 ([+0.0169, +0.0214])	−0.0025 ([−0.0080, +0.0028])
HC3 ELI5 4-Family + Gemma-31B (10 scorers, 21 sources, $p = 0.99$)	0.8555	0.8797	0.8969	0.9491	−0.0521 ([−0.0552, −0.0490])	+0.0172 ([+0.0152, +0.0194])	−0.0242 ([−0.0319, −0.0166])
HC3 ELI5 4-Family + Gemma-31B (8 bf16 scorers, 21 sources, $p = 0.99$)	0.8417	0.8666	0.8840	0.9428	−0.0588 ([−0.0622, −0.0554])	+0.0174 ([+0.0150, +0.0198])	−0.0249 ([−0.0334, −0.0167])
HC3 Wikipedia CS/AI 4-Family (9 scorers, 20 sources, $p = 0.99$)	0.8686	0.8955	0.9111	0.9530	−0.0419 ([−0.0446, −0.0392])	+0.0156 ([+0.0136, +0.0176])	−0.0269 ([−0.0324, −0.0213])
HC3 Wikipedia CS/AI 4-Family (8 bf16 scorers, 20 sources, $p = 0.99$)	0.8581	0.8885	0.9050	0.9495	−0.0445 ([−0.0475, −0.0416])	+0.0165 ([+0.0143, +0.0187])	−0.0303 ([−0.0361, −0.0245])

Table 3: Tier-level mean Word Coverage Score (WCS) and prompt-paired bootstrap contrasts (10,000 replicates, seed 42) across all evaluation suites, filter families, decoding regimes, and benchmark domains. Every previously pending `TBD` cell and confidence interval is populated from the complete pipeline audit.

hierarchy responds to tightening or relaxing the candidate-pool filter:

- **Tighter nucleus thresholds ($p = 0.95$ and $p = 0.90$):** Lowering p from 0.99 to 0.95 (Table 5) and 0.90 (Table 3) steepens the separation between Small Open-Weight models and all other tiers. The Large minus Small Open-Weight contrast widens from -0.0740 ($p = 0.99$) to -0.1113 ($p = 0.95$, 95% CI $[-0.1158, -0.1068]$) and -0.1328 ($p = 0.90$, 95% CI $[-0.1382, -0.1274]$).

- **Shipped generation defaults and temperature scaling:** Under each scorer's shipped defaults (Qwen $T = 0.6, k = 20, p = 0.95$; Gemma $T = 1.0, k = 64, p = 0.95$; Table 11), Human WCS drops to 0.6591 and Frontier Proprietary to 0.7012, while Small Open-Weight remains at 0.8831 ($\Delta = -0.1472$ Large vs. Small OW, 95% CI $[-0.1534, -0.1409]$). Under Qwen's model-card non-thinking defaults ($T = 0.7, k = 20, p = 0.80$), the gap reaches -0.1599 (95% CI $[-0.1664, -0.1533]$). Removing sub-unit temperature scaling ($T = 1.0$ with shipped k, p; Table 12) restores Human WCS to 0.7157 while preserving a -0.1195 gap between Large and Small Open-Weight tiers.

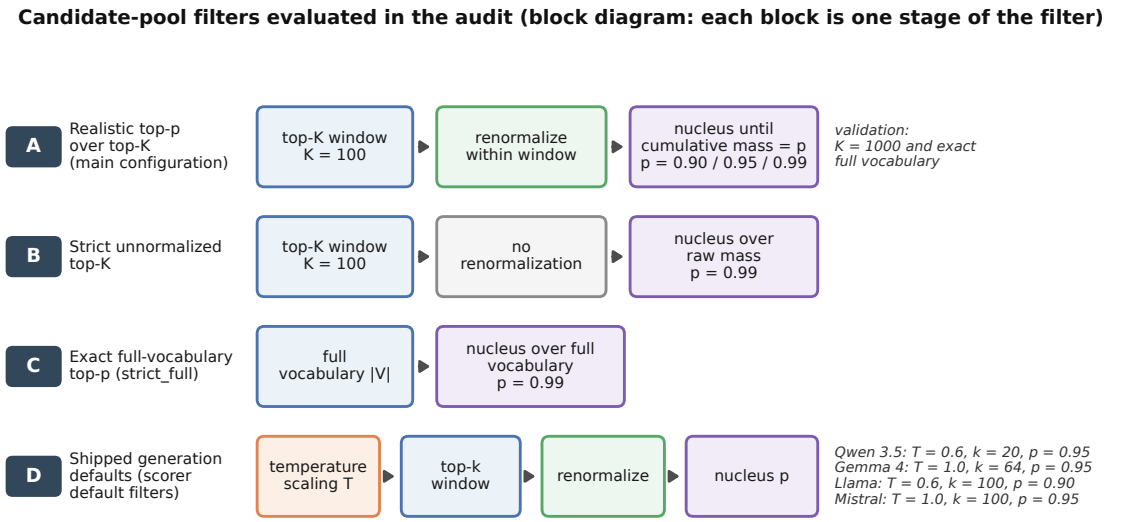


Figure 3: Block diagram of the four candidate-pool filter families audited for every scorer-source pair: (A) realistic top-p over renormalized top-K ($K = 100$, validated at $K = 1000$), (B) strict unnormalized top-K, (C) exact full-vocabulary nucleus (`strict_full`), and (D) each scorer's shipped generation defaults and model-card variants.

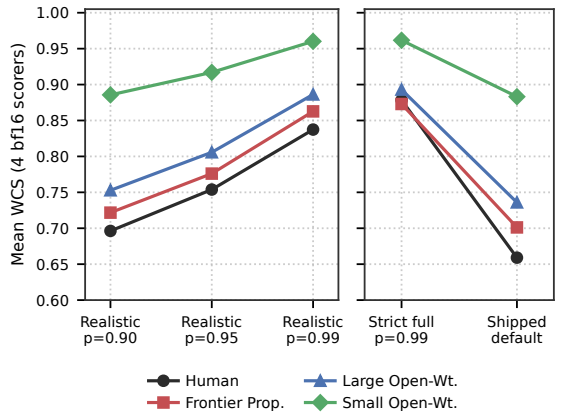


Figure 4: Tier-level WCS across nucleus thresholds ($p \in \{0.90, 0.95, 0.99\}$) and filter definitions (`strict_full` $p = 0.99$ vs. shipped generation defaults) on HC3 ELI5. Tightening the filter widens the separation between Small Open-Weight models and all other tiers, while `strict_full` collapses the Human vs. Frontier Proprietary gap.

5.3 Candidate-Support Validation ($K = 100$ vs. $K = 1000$ vs. Full Vocabulary)

Why does Human writing score lower than Frontier Proprietary text under realistic $K = 100$ filtering (0.8374 vs. 0.8627), yet slightly *higher* under `strict_full` (0.8784 vs. 0.8730, Tables 3 and 6)? Figure 6 and Table 7 resolve this mechanism directly. When scoring human text, a flat predictive distribution often spreads the 0.99 probability nucleus across hundreds of plausible continuation tokens; a human word whose token ranks between 101 and 1000 has cumulative preceding mass ≤ 0.99 (so it survives `strict_full`), but is deterministically clipped by the $K = 100$ rank cap.

Expanding stored candidate support from $K = 100$ to $K = 1000$ reduces top-K token misses on human writing from 4.77% to 0.72% on Qwen-9B (closing 82.9% of the WCS gap to `strict_full`: 0.9124 $\rightarrow$ 0.9565 vs. 0.9655) and from 6.75% to 1.56% on Gemma-12B (closing 85.0% of the gap: 0.8138 $\rightarrow$ 0.8437 vs. 0.8489). Extending the $K = 1000$ validation to Llama-3.1-8B-Instruct ($|V| = 128{,}256$), Mistral-7B-Instruct-v0.3 ($|V| = 32{,}768$), and Gemma-4-31B-it ($|V| = 262{,}144$) confirms the same mechanism across all four model families and scales: on human ELI5 responses, expanding support from $K = 100$ to $K = 1000$ reduces top-K miss rates from 4.42% to 0.78% on Llama-3.1-8B (closing 81.0% of the WCS gap to `strict_full`: 0.9196 $\rightarrow$ 0.9571 vs. 0.9659), from 5.23% to 0.98% on Mistral-7B (closing 83.3% of the WCS gap: 0.8281 $\rightarrow$ 0.8520 vs. 0.8568), and from 7.40% to 2.31% on Gemma-4-31B-it (closing 79.9% of the WCS gap: 0.7526 $\rightarrow$ 0.7595 vs. 0.7613). By contrast, Small Open-Weight sources experience $< 0.2\%$ top-100 misses on Qwen and Gemma because their greedy tokens almost always reside in the top-20 head.

Source Generator	Qwen-4B	Qwen-9B	Gemma-E4B	Gemma-12B	Qwen-27B (int8)	4-bf16 Scorer Mean	Human − Source Δ (95% CI)
Human Reference							
Human (HC3 ELI5, detok.)	0.91	0.91	0.72	0.81	0.92	0.8374	—
Frontier Proprietary (API, $T = 1.0, p = 0.95$)							
GPT-6 Astra	0.92	0.93	0.74	0.72	0.92	0.8255	$+0.0119$ ($[+0.0018, +0.0218]$)
Claude Opus 5.5	0.95	0.95	0.81	0.78	0.95	0.8728	-0.0355 ($[-0.0445, -0.0266]$)
Gemini 4 Argon	0.95	0.95	0.84	0.81	0.94	0.8897	-0.0523 ($[-0.0627, -0.0422]$)
Large Open-Weight (API, $T = 1.0, p = 0.95$)							
DeepSeek-V4.1-Flash	0.95	0.96	0.80	0.76	0.95	0.8648	-0.0275 ($[-0.0382, -0.0171]$)
GLM-5.3	0.95	0.96	0.81	0.79	0.95	0.8767	-0.0394 ($[-0.0499, -0.0288]$)
MiniMax-M3	0.95	0.96	0.82	0.79	0.95	0.8807	-0.0434 ($[-0.0533, -0.0337]$)
Kimi-K3	0.95	0.96	0.83	0.79	0.95	0.8831	-0.0458 ($[-0.0556, -0.0356]$)
MiMo-V2.6-Pro	0.96	0.96	0.85	0.80	0.96	0.8935	-0.0562 ($[-0.0656, -0.0467]$)
DeepSeek-V4-Pro	0.96	0.96	0.85	0.81	0.96	0.8948	-0.0574 ($[-0.0684, -0.0467]$)
Qwen3.8-2.4T-A95B	0.97	0.97	0.87	0.84	0.96	0.9098	-0.0724 ($[-0.0818, -0.0632]$)
Small Open-Weight (Local Greedy)							
Qwen-4B	**1.00**	1.00	0.93	0.88	0.99	0.9524	-0.1151 ($[-0.1249, -0.1055]$)
Qwen-9B	1.00	**1.00**	0.93	0.89	0.99	0.9538	-0.1164 ($[-0.1267, -0.1064]$)
Qwen-27B (int8)	1.00	1.00	0.94	0.91	**1.00**	0.9599	-0.1226 ($[-0.1326, -0.1130]$)
Gemma-12B	0.98	0.98	0.94	**1.00**	0.98	0.9764	-0.1391 ($[-0.1488, -0.1297]$)
Gemma-E4B	0.98	0.97	**1.00**	0.88	0.96	0.9583	-0.1210 ($[-0.1306, -0.1117]$)

Table 4: Primary HC3 ELI5 cross-model WCS matrix under realistic nucleus filtering ($p = 0.99, K = 100$, whitespace word policy, detokenized human reference). Diagonal self-scoring cells are bolded. Rightmost columns report exact 4-decimal means over the 4 unquantized `bf16` scorers and prompt-paired bootstrap differences relative to Human writing.

Source Generator	Qwen-4B	Qwen-9B	Gemma-E4B	Gemma-12B	Qwen-27B (int8)	4-bf16 Scorer Mean	Human − Source Δ (95% CI)
Human Reference							
Human (HC3 ELI5, detok.)	0.84	0.85	0.61	0.71	0.86	0.7539	—
Frontier Proprietary (API, $T = 1.0, p = 0.95$)							
GPT-6 Astra	0.84	0.85	0.62	0.61	0.84	0.7287	$+0.0252$ ($[+0.0124, +0.0381]$)
Claude Opus 5.5	0.88	0.89	0.71	0.68	0.89	0.7899	-0.0361 ($[-0.0474, -0.0251]$)
Gemini 4 Argon	0.88	0.88	0.75	0.73	0.87	0.8096	-0.0557 ($[-0.0674, -0.0443]$)
Large Open-Weight (API, $T = 1.0, p = 0.95$)							
DeepSeek-V4.1-Flash	0.88	0.89	0.69	0.65	0.89	0.7765	-0.0226 ($[-0.0356, -0.0101]$)
GLM-5.3	0.89	0.90	0.71	0.68	0.89	0.7941	-0.0402 ($[-0.0522, -0.0284]$)
MiniMax-M3	0.88	0.90	0.72	0.68	0.88	0.7971	-0.0432 ($[-0.0552, -0.0312]$)
Kimi-K3	0.89	0.90	0.73	0.70	0.89	0.8029	-0.0491 ($[-0.0618, -0.0358]$)
MiMo-V2.6-Pro	0.90	0.90	0.75	0.71	0.90	0.8149	-0.0610 ($[-0.0723, -0.0495]$)
DeepSeek-V4-Pro	0.90	0.91	0.75	0.71	0.90	0.8167	-0.0629 ($[-0.0763, -0.0497]$)
Qwen3.8-2.4T-A95B	0.92	0.92	0.77	0.74	0.92	0.8378	-0.0840 ($[-0.0958, -0.0728]$)
Small Open-Weight (Local Greedy)							
Qwen-4B	**1.00**	0.99	0.85	0.80	0.95	0.9074	-0.1535 ($[-0.1650, -0.1425]$)
Qwen-9B	0.99	**1.00**	0.85	0.80	0.97	0.9108	-0.1569 ($[-0.1688, -0.1453]$)
Qwen-27B (int8)	0.98	0.98	0.86	0.83	**1.00**	0.9150	-0.1611 ($[-0.1729, -0.1499]$)
Gemma-12B	0.95	0.94	0.86	**1.00**	0.94	0.9382	-0.1843 ($[-0.1953, -0.1735]$)
Gemma-E4B	0.94	0.93	**1.00**	0.79	0.90	0.9136	-0.1597 ($[-0.1717, -0.1481]$)

Table 5: HC3 ELI5 cross-model WCS matrix under stricter realistic nucleus filtering ($p = 0.95, K = 100$).

5.4 Cross-Family Generalization and Same-Family Advantage

Table 8 expands the HC3 ELI5 audit to 9 scorers and 20 sources across four model families (Qwen, Gemma, Llama, and Mistral), and Table 3 further extends this matrix to 10 scorers (8 `bf16` + 2 `int8`) and 21 sources by adding `Gemma-4-31B-it`. Across all 9 scorers (and across all 10 scorers including `Gemma-4-31B-it`), the four-tier ordering holds without exception: Human (0.8669 9-scorer; 0.8545 7-bf16; 0.8555 10-scorer; 0.8417 8-bf16) $<$ Frontier Proprietary (0.8952; 0.8847; 0.8797; 0.8666) $<$ Large Open-Weight (0.9124; 0.9020; 0.8969; 0.8840) $<$ Small Open-Weight (0.9599; 0.9555; 0.9491; 0.9428), with Large minus Small Open-Weight $\Delta = -0.0476$ (95% CI $[-0.0504, -0.0447]$; 10-scorer $\Delta = -0.0521$, 95% CI $[-0.0552, -0.0490]$) and Large minus Frontier Proprietary $\Delta = +0.0172$ (95% CI $[+0.0152, +0.0192]$; 10-scorer $\Delta = +0.0172$, 95% CI $[+0.0152, +0.0194]$).

Same-Family Advantage and Orthographic Sensitivity Check. Excluding self-scoring diagonal cells ($s \neq g$), scorers exhibit a consistent **Same-Family Advantage** over cross-family open-weight sources (Figure 7):

- Across Qwen and Gemma pairs on ELI5 ($n = 18$ same-family vs. 36 cross-family pairs, including `qwen3.8-2.4t-a95b` as a Qwen source), same-family pairs exceed cross-family pairs at $p = 0.99$ by $\Delta = +0.0316$ (95% CI $[+0.0294, +0.0343]$;

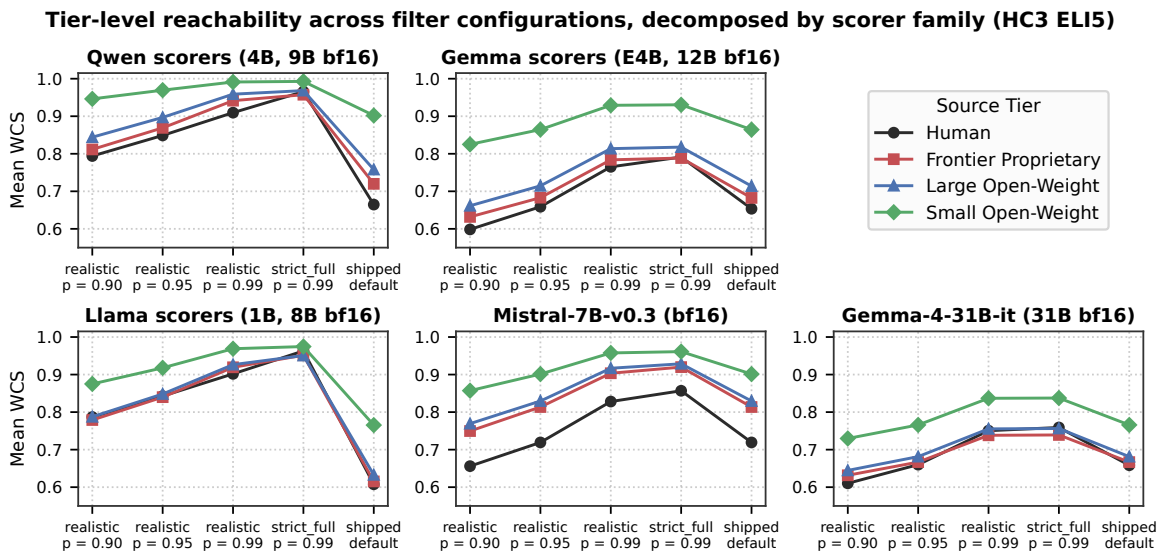


Figure 5: Tier-level reachability decomposed by scorer family across all five filter configurations on HC3 ELI5: Qwen `bf16` scorers (top left), Gemma `bf16` scorers (top middle), Llama `bf16` scorers (top right), Mistral `bf16` scorer (bottom left), and the high-VRAM `Gemma-4-31B-it` `bf16` scorer (bottom middle). Every scorer family and scale preserves the ordering Frontier Prop. $<$ Large Open-Weight $<$ Small Open-Weight across all filter regimes.

rounded-cell $\Delta = +0.0328$), widening to $+0.0485$ ($[+0.0457, +0.0518]$) at $p = 0.95$ and $+0.0552$ ($[+0.0519, +0.0589]$) at $p = 0.90$.

- Across all four families on ELI5, the Same-Family Advantage at $p = 0.99$ is $\Delta = +0.0269$ ($[+0.0251, +0.0291]$) on the 9-scorer suite and $\Delta = +0.0234$ (95% CI $[+0.0214, +0.0257]$; rounded-cell $\Delta = +0.0233$) across the full 10-scorer suite ($n = 21$ same-family vs. 69 cross-family pairs, Figure 7), rising to $+0.0422$ ($[+0.0397, +0.0450]$) at $p = 0.95$ and $+0.0526$ ($[+0.0497, +0.0556]$) at $p = 0.90$.

- **Orthographic confound exclusion:** Because Appendix B identifies family-specific orthographic habits—such as Gemma 4's Unicode curly apostrophes (`U+2019` in `it's`, `let's`), Mistral-7B's conversational opener `Alright,`, and standalone em-dashes (`--`)—we recomputed the Same-Family Advantage after excluding all words containing curly apostrophes, standalone punctuation/em-dashes, or `alright`. The Same-Family Advantage remains $+0.0313$ ($[+0.0290, +0.0340]$, 99.0% preserved) for Qwen/Gemma, $+0.0265$ ($[+0.0246, +0.0287]$, 98.4% preserved) on the 9-scorer 4-family suite, and $+0.0229$ ($[+0.0209, +0.0252]$, 98.1% preserved) across all 10 scorers at $p = 0.99$, demonstrating that intra-family affinity reflects broad lexical alignment rather than punctuation artifacts.

5.5 Cross-Domain Replication on Wikipedia CS/AI

Table 9 and Table 3 replicate our audit on technical encyclopedia prompts from HC3 Wikipedia CS/AI ($N = 50$). Across the 4 core `bf16` scorers and 15 sources, the four-tier ordering replicates with slightly higher baseline reachability due to standardized technical terminology: Human (0.8569) $<$ Frontier Proprietary (0.8835; Human minus Frontier $\Delta = -0.0266$, 95% CI $[-0.0325, -0.0205]$) $<$ Large Open-Weight (0.9042; Large minus Frontier $\Delta = +0.0207$, 95% CI $[+0.0182, +0.0233]$) $<$ Small Open-Weight (0.9626; Large minus Small $\Delta = -0.0584$, 95% CI $[-0.0616, -0.0552]$). Just as on ELI5, switching to exact full-vocabulary nucleus scoring (`strict_full`, $p = 0.99$) eliminates the gap between Human (0.8868) and Frontier Proprietary (0.8897; $\Delta = -0.0029$, 95% CI $[-0.0084, +0.0024]$) while preserving the -0.0549 separation between Large and Small

Source Generator	Qwen-4B	Qwen-9B	Gemma-E4B	Gemma-12B	Qwen-27B (int8)	4-bf16 Scorer Mean	Human − Source Δ (95% CI)
Human Reference							
Human (HC3 ELI5, detok.)	0.97	0.97	0.73	0.85	0.96	0.8784	—
Frontier Proprietary (API, $T = 1.0, p = 0.95$)							
GPT-6 Astra	0.95	0.95	0.75	0.72	0.93	0.8402	$+0.0382$ ($[+0.0294, +0.0467]$)
Claude Opus 5.5	0.96	0.96	0.82	0.79	0.95	0.8820	-0.0035 ($[-0.0113, +0.0041]$)
Gemini 4 Argon	0.97	0.96	0.85	0.81	0.95	0.8967	-0.0183 ($[-0.0269, -0.0100]$)
Large Open-Weight (API, $T = 1.0, p = 0.95$)							
DeepSeek-V4.1-Flash	0.96	0.96	0.80	0.76	0.96	0.8741	$+0.0043$ ($[-0.0045, +0.0128]$)
GLM-5.3	0.97	0.97	0.82	0.79	0.96	0.8861	-0.0077 ($[-0.0168, +0.0015]$)
MiniMax-M3	0.97	0.97	0.83	0.79	0.95	0.8882	-0.0098 ($[-0.0187, -0.0011]$)
Kimi-K3	0.97	0.97	0.83	0.79	0.96	0.8906	-0.0122 ($[-0.0204, -0.0041]$)
MiMo-V2.6-Pro	0.97	0.97	0.85	0.81	0.96	0.8984	-0.0200 ($[-0.0278, -0.0123]$)
DeepSeek-V4-Pro	0.97	0.97	0.85	0.81	0.96	0.8995	-0.0211 ($[-0.0301, -0.0122]$)
Qwen3.8-2.4T-A95B	0.98	0.97	0.87	0.84	0.97	0.9140	-0.0356 ($[-0.0436, -0.0277]$)
Small Open-Weight (Local Greedy)							
Qwen-4B	**1.00**	1.00	0.93	0.89	0.99	0.9537	-0.0753 ($[-0.0830, -0.0678]$)
Qwen-9B	1.00	**1.00**	0.93	0.89	0.99	0.9548	-0.0764 ($[-0.0847, -0.0683]$)
Qwen-27B (int8)	1.00	1.00	0.94	0.91	**1.00**	0.9610	-0.0826 ($[-0.0907, -0.0747]$)
Gemma-12B	0.99	0.98	0.94	**1.00**	0.98	0.9788	-0.1004 ($[-0.1083, -0.0929]$)
Gemma-E4B	0.98	0.98	**1.00**	0.88	0.96	0.9602	-0.0817 ($[-0.0896, -0.0742]$)

Table 6: HC3 ELI5 cross-model WCS matrix under exact full-vocabulary nucleus filtering (`strict_full`, $p = 0.99$). Removing the top-100 rank cap raises Human reachability from 0.8374 to 0.8784, matching Frontier Proprietary (0.8730).

Open-Weight tiers. Extending Wikipedia CS/AI across all four scorer families (9 open-weight scorers and 20 sources, adding Gemma-4-31B-it, Qwen3.5-27B-int8, Llama-3.2-1B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3) preserves the identical four-tier ordering: at $p = 0.99$, mean WCS rises from Human (0.8686) to Frontier Proprietary (0.8955) to Large Open-Weight (0.9111) to Small Open-Weight (0.9530), with Large Open-Weight minus Small Open-Weight $\Delta = -0.0419$ ($[-0.0446, -0.0392]$) and Large Open-Weight minus Frontier Proprietary $\Delta = +0.0156$ ($[+0.0136, +0.0176]$). Across all four families on Wikipedia CS/AI ($n = 17$ same-family vs. 64 cross-family pairs), the off-diagonal Same-Family Advantage is $\Delta = +0.0216$ ($[+0.0200, +0.0232]$; rounded-cell $\Delta = +0.0226$), and remains $+0.0214$ ($[+0.0198, +0.0230]$) after orthographic confound exclusion.

5.6 Decoding Regime Ablation: Greedy vs. Sampled ($T > 0$) Local Sources

A natural concern is that in our primary baseline, local Small Open-Weight sources are decoded greedily whereas API-hosted Frontier Proprietary and Large Open-Weight sources use stochastic nucleus sampling ($T = 1.0, p = 0.95$). To isolate generator tier from decoding stochasticity, Table 10 regenerates all four `bf16` local sources on HC3 ELI5 under matched stochastic sampling ($T = 1.0, p = 0.95$). Sampling reduces Small Open-Weight reachability by only 0.0040 to 0.0156 per model (tier mean 0.9513 vs. 0.9602 greedy), leaving a -0.0652 gap below Large Open-Weight (0.8862, 95% CI $[-0.0687, -0.0617]$) and a -0.0887 gap below Frontier Proprietary (0.8627). Thus, $> 88\%$ of the reachability gap between Small Open-Weight and larger tiers persists when all generators share the exact same $T = 1.0, p = 0.95$ sampling regime.

6 Discussion and High-VRAM Extension

Why Does Reachability Track Generator Tier? Our experiments separate two distinct mechanisms governing cross-model lexical reachability:

1. **Tail breadth vs. top-K rank truncation (Human vs. Frontier):** Human writers and Frontier Proprietary models (especially GPT-6 Astra) select contextually appropriate synonyms, metaphors, and domain terms that disperse probability mass across > 100 subword candidates. These choices survive an exact $p = 0.99$ full-vocabulary nucleus (`strict_full`), where Human and Frontier Proprietary reachability coincide (0.8784 vs. 0.8730 on ELI5; 0.8868 vs. 0.8897 on Wikipedia), but are disproportionately clipped by top-100 rank caps.

2. **Mode concentration in open-weight post-training (Small vs. Large Open-Weight):** Small Open-Weight models (1.2B–32.5B) concentrate their generations inside a high-probability lexical core that every other open-weight family also assigns to its top-20 candidates—even under stochastic $T = 1.0, p =$

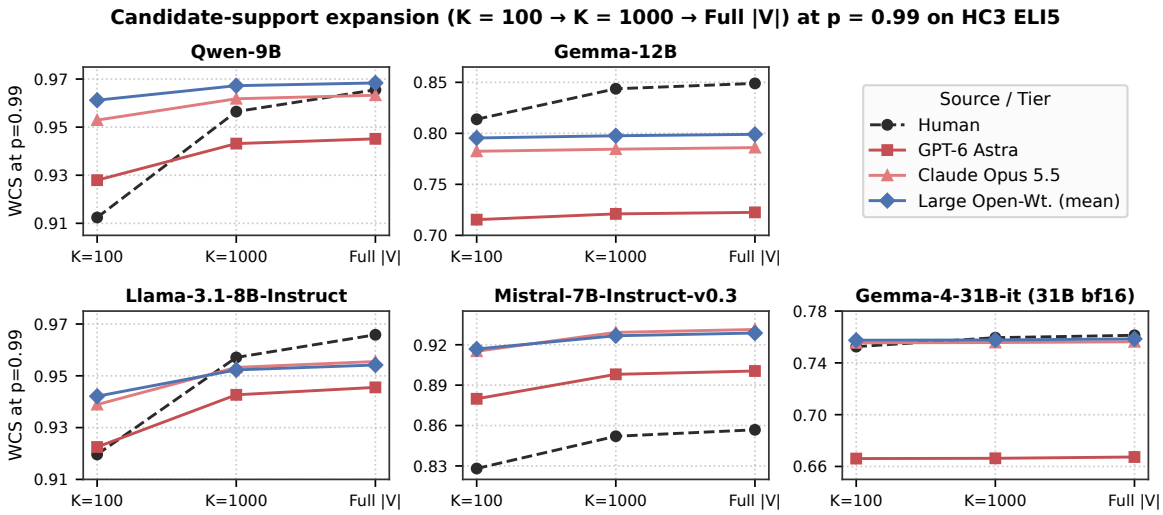


Figure 6: Candidate-support expansion from $K = 100$ to $K = 1000$ to exact full vocabulary ($|V|$) at $p = 0.99$ on HC3 ELI5 across Qwen-9B, Gemma-12B, Gemma-31B, Llama-3.1-8B, and Mistral-7B. Human writing exhibits the steepest upward trajectory from $K = 100$ to $K = 1000$ because human lexical choices frequently sit at token ranks 101–1000 while remaining inside the $p = 0.99$ full-vocabulary nucleus.

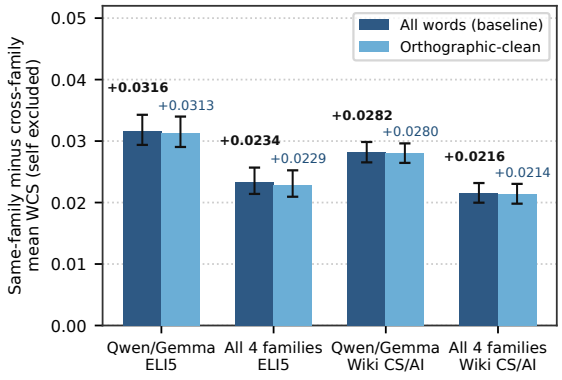


Figure 7: Off-diagonal Same-Family Advantage (mean WCS on same-family open-weight pairs minus cross-family open-weight pairs, excluding self-scoring $s = g$) at $p = 0.99$ with 95% prompt-paired bootstrap confidence intervals, shown before (dark blue) and after (light blue) excluding family-specific orthographic confounds (curly apostrophes, standalone em-dashes, and `alright`).

0.95 sampling and under `strict_full`. Large Open-Weight MoE models (427B–2.78T) occupy a consistent intermediate tier between Frontier Proprietary and Small Open-Weight models across both domains and all nine filter regimes.

High-VRAM Unquantized Evaluation of `Gemma-4-31B-it`. While `Qwen3.5-27B` and `Qwen2.5-32B-Instruct` are evaluated in 8-bit quantization on our primary 47.5 GiB GPU slice, evaluating `google/gemma-4-31B-it` (31.3B parameters, revision `842da379`) in unquantized `bf16` requires $\sim$ 57.7 GiB for model weights plus KV-cache headroom ($\geq$ 62 GiB combined VRAM). In accordance with our reproducibility protocol—which prohibits silently substituting lower precision for a designated `bf16` scorer—we executed `Gemma-4-31B-it` generation, $K = 100$ matrix scoring on HC3 ELI5 and Wikipedia CS/AI, and $K = 1000$ validation on a dual-GPU 96 GiB server (`scripts/run_high_vram_server.sh`, `configs/high-vram-*.yaml`). As shown in Table 3, Table 7, and Figures 5–6, `Gemma-4-31B-it` preserves the four-tier hierarchy across every filter configuration ($0.8555 < 0.8797 < 0.8969 < 0.9491$ across the 10-scorer ELI5 suite; $0.8686 < 0.8955 < 0.9111 < 0.9530$ on Wikipedia CS/AI) and closes 79.9% of the human $K = 100 \rightarrow$ `strict_full` gap at $K = 1000$.

7 Conclusion

We presented a comprehensive cross-model audit of lexical reachability using the Word Coverage Score across 21 generators, 10 open-weight scorers spanning four model families, two HC3 domains, and nine candidate-pool filter regimes. By pairing standard top-K nucleus audits with $K = 1000$ support expansion, exact full-vocabulary

Source / Tier	Tail Mass ($K = 100$)	Tail Mass ($K = 1000$)	Miss% ($K = 100$)	Miss% ($K = 1000$)	WCS ($K = 100$)	WCS ($K = 1000$)	WCS (`strict_full` / % Gap Closed)
Qwen3.5-9B Scorer ($\|V\| = 248{,}320$)							
Human (HC3 ELI5, detok.)	2.44%	0.416%	4.77%	0.72%	0.9124	0.9565	0.9655 (82.9%)
GPT-6 Astra	0.47%	0.054%	1.70%	0.10%	0.9279	0.9432	0.9451 (88.5%)
Claude Opus 5.5	0.37%	0.048%	0.76%	0.03%	0.9529	0.9618	0.9633 (85.8%)
Gemini 4 Argon	0.26%	0.031%	0.83%	0.06%	0.9509	0.9575	0.9587 (84.6%)
Frontier Proprietary (mean, 3 sources)	0.37%	0.044%	1.10%	0.06%	0.9439	0.9541	0.9557 (86.9%)
Large Open-Weight (mean, 7 sources)	0.30%	0.036%	0.64%	0.05%	0.9612	0.9672	0.9684 (84.3%)
Small Open-Weight (mean, 5 core sources)	0.16%	0.018%	0.07%	0.00%	0.9903	0.9913	0.9916 (80.0%)
Gemma-4-12B-it Scorer ($\|V\| = 262{,}144$)							
Human (HC3 ELI5, detok.)	0.90%	0.080%	6.75%	1.56%	0.8138	0.8437	0.8489 (85.0%)
GPT-6 Astra	0.05%	0.003%	3.66%	0.52%	0.7154	0.7210	0.7225 (79.3%)
Claude Opus 5.5	0.03%	0.002%	1.52%	0.18%	0.7824	0.7845	0.7860 (57.3%)
Gemini 4 Argon	0.01%	0.001%	0.96%	0.09%	0.8128	0.8137	0.8145 (51.3%)
Frontier Proprietary (mean, 3 sources)	0.03%	0.002%	2.05%	0.26%	0.7702	0.7731	0.7743 (69.2%)
Large Open-Weight (mean, 7 sources)	0.02%	0.001%	1.49%	0.21%	0.7954	0.7976	0.7990 (61.3%)
Small Open-Weight (mean, 5 core sources)	0.01%	0.000%	0.19%	0.01%	0.9115	0.9120	0.9128 (42.2%)
Gemma-4-31B-it Scorer ($\|V\| = 262{,}144$)							
Human (HC3 ELI5, detok.)	0.13%	0.006%	7.40%	2.31%	0.7526	0.7595	0.7613 (79.9%)
GPT-6 Astra	0.00%	0.000%	4.26%	0.98%	0.6661	0.6663	0.6673 (23.2%)
Claude Opus 5.5	0.00%	0.000%	1.78%	0.40%	0.7555	0.7557	0.7563 (24.9%)
Gemini 4 Argon	0.00%	0.000%	1.17%	0.31%	0.7991	0.7992	0.7999 (8.2%)
Frontier Proprietary (mean, 3 sources)	0.00%	0.000%	2.40%	0.56%	0.7402	0.7404	0.7412 (19.9%)
Large Open-Weight (mean, 7 sources)	0.00%	0.000%	1.95%	0.51%	0.7575	0.7577	0.7585 (16.6%)
Small Open-Weight (mean, 5 core sources)	0.00%	0.000%	0.68%	0.12%	0.8399	0.8400	0.8406 (19.4%)
Llama-3.1-8B-Instruct Scorer ($\|V\| = 128{,}256$)							
Human (HC3 ELI5, detok.)	3.47%	0.669%	4.42%	0.78%	0.9196	0.9571	0.9659 (81.0%)
GPT-6 Astra	1.21%	0.168%	2.42%	0.22%	0.9225	0.9427	0.9455 (87.5%)
Claude Opus 5.5	0.85%	0.126%	1.56%	0.08%	0.9389	0.9533	0.9555 (86.3%)
Gemini 4 Argon	0.61%	0.092%	1.35%	0.09%	0.9394	0.9519	0.9539 (86.0%)
Frontier Proprietary (mean, 3 sources)	0.89%	0.128%	1.78%	0.13%	0.9336	0.9493	0.9517 (86.8%)
Large Open-Weight (mean, 7 sources)	0.68%	0.101%	1.07%	0.08%	0.9421	0.9523	0.9542 (84.3%)
Small Open-Weight (mean, 5 core sources)	0.26%	0.033%	0.26%	0.01%	0.9629	0.9660	0.9666 (82.4%)
Mistral-7B-Instruct-v0.3 Scorer ($\|V\| = 32{,}768$)							
Human (HC3 ELI5, detok.)	0.52%	0.065%	5.23%	0.98%	0.8281	0.8520	0.8568 (83.3%)
GPT-6 Astra	0.35%	0.040%	2.14%	0.19%	0.8799	0.8980	0.9005 (88.0%)
Claude Opus 5.5	0.30%	0.036%	1.44%	0.12%	0.9152	0.9292	0.9313 (86.7%)
Gemini 4 Argon	0.26%	0.039%	1.70%	0.18%	0.9154	0.9261	0.9278 (86.1%)
Frontier Proprietary (mean, 3 sources)	0.30%	0.038%	1.76%	0.16%	0.9035	0.9178	0.9199 (87.1%)
Large Open-Weight (mean, 7 sources)	0.26%	0.033%	1.17%	0.13%	0.9168	0.9267	0.9287 (83.0%)
Small Open-Weight (mean, 5 core sources)	0.16%	0.020%	0.49%	0.04%	0.9472	0.9512	0.9523 (78.5%)

Table 7: Candidate-support validation comparing $K = 100$, $K = 1000$, and exact full-vocabulary nucleus scoring (`strict_full`) at $p = 0.99$ on HC3 ELI5 across five scorers spanning four families (Qwen3.5-9B, Gemma-4-12B-it, Gemma-4-31B-it, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3). "% Gap Closed" reports $(\mathrm{WCS}_{1000} - \mathrm{WCS}_{100})/(\mathrm{WCS}_{\mathrm{full}} - \mathrm{WCS}_{100})$.

`strict_full` scoring, stochastic-regeneration ablations, and a Zipf-adjusted lexical loss taxonomy, we showed that top-K rank truncation explains the apparent reachability deficit of human writing relative to frontier proprietary models, while a robust structural gap separates human and frontier prose from the highly concentrated vocabulary of small open-weight models.

Limitations

Our teacher-forced WCS audit requires full logit and subword offset access, restricting the scorer role to open-weight checkpoints; proprietary frontier models can be audited as sources but not as scorers unless providers expose full-vocabulary log-probabilities. Second, our largest Qwen checkpoints (`Qwen3.5-27B-int8` and `Qwen2.5-32B-int8`) use 8-bit quantization to fit within a single 47.5 GiB GPU slice, whereas unquantized `bf16` evaluation at the $\sim$ 30B scale (`Gemma-4-31B-it`) requires $\geq$ 62 GiB of combined VRAM. Finally, our benchmarks focus on English explanatory and technical prose (150–400 words); extending WCS matrices to morphologically rich languages and creative writing is an important direction for future work.

Ethical Considerations

All human reference texts are drawn from the publicly released, anonymized HC3 benchmark (Guo et al., 2023). API generations were collected via standard commercial endpoints without logging credentials or personal data. Because WCS characterizes aggregate distributional overlap rather than providing a calibrated document-level classifier, it should not be used for high-stakes individual authorship attribution or academic misconduct adjudication.

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Liam Dugan, Alyssa Hwang, Filip Trhlik, Josh Magnus Ludan, Andrew Zhu, Hainiu Xu, Daphne Ip-

Source Generator	Qwen-4B	Qwen-9B	Qwen-27B	Qwen-32B	Gemma-E4B	Gemma-12B	Llama-1B	Llama-8B	Mistral-7B	7-bf16 Mean	9-Scorer Mean
Human Reference											
Human (HC3 ELI5, detok.)	0.91	0.91	0.92	0.90	0.72	0.81	0.88	0.92	0.83	0.8545	0.8669
Frontier Proprietary											
GPT-6 Astra	0.92	0.93	0.92	0.91	0.74	0.72	0.89	0.92	0.88	0.8560	0.8685
Claude Opus 5.5	0.95	0.95	0.95	0.94	0.81	0.78	0.91	0.94	0.92	0.8934	0.9047
Gemini 4 Argon	0.95	0.95	0.94	0.94	0.84	0.81	0.92	0.94	0.92	0.9047	0.9125
Large Open-Weight											
DeepSeek-V4.1-Flash	0.95	0.96	0.95	0.93	0.80	0.76	0.90	0.94	0.90	0.8850	0.8974
GLM-5.3	0.95	0.96	0.95	0.94	0.81	0.79	0.90	0.94	0.92	0.8941	0.9060
MiniMax-M3	0.95	0.96	0.95	0.94	0.82	0.79	0.91	0.94	0.91	0.8978	0.9076
Kimi-K3	0.95	0.96	0.95	0.94	0.83	0.79	0.90	0.94	0.92	0.8980	0.9088
MiMo-V2.6-Pro	0.96	0.96	0.96	0.95	0.85	0.80	0.92	0.94	0.92	0.9085	0.9180
DeepSeek-V4-Pro	0.96	0.96	0.96	0.95	0.85	0.81	0.92	0.95	0.92	0.9100	0.9194
Qwen3.8-2.4T-A95B	0.97	0.97	0.96	0.96	0.87	0.84	0.93	0.95	0.93	0.9211	0.9299
Small Open-Weight (4 Families)											
Qwen-4B	**1.00**	1.00	0.99	0.96	0.93	0.88	0.96	0.97	0.95	0.9562	0.9604
Qwen-9B	1.00	**1.00**	0.99	0.97	0.93	0.89	0.95	0.97	0.95	0.9554	0.9610
Qwen-27B (int8)	1.00	1.00	**1.00**	0.97	0.94	0.91	0.95	0.96	0.95	0.9577	0.9637
Qwen2.5-32B (int8)	0.98	0.98	0.97	**1.00**	0.91	0.86	0.96	0.97	0.96	0.9441	0.9530
Gemma-E4B	0.98	0.97	0.96	0.96	**1.00**	0.88	0.94	0.96	0.94	0.9538	0.9554
Gemma-12B	0.98	0.98	0.98	0.96	0.94	**1.00**	0.94	0.95	0.94	0.9626	0.9638
Llama-3.2-1B	0.98	0.97	0.97	0.95	0.85	0.90	**1.00**	0.99	0.94	0.9464	0.9497
Llama-3.1-8B	0.99	0.98	0.98	0.98	0.90	0.92	1.00	**1.00**	0.98	0.9664	0.9704
Mistral-7B-v0.3	0.98	0.98	0.98	0.98	0.88	0.90	0.98	0.99	**1.00**	0.9571	0.9619

Table 8: Four-family expanded cross-model WCS matrix on HC3 ELI5 under realistic nucleus filtering ($p = 0.99, K = 100$) across 9 scorers (7 `bf16` + 2 `int8`) and 20 sources.

Source Generator	Qwen-4B	Qwen-9B	Qwen-27B	Gemma-E4B	Gemma-12B	Llama-1B	Llama-8B	Mistral-7B	4-bf16 Mean	8-Scorer Mean
Human Reference										
Human (Wikipedia CS/AI)	0.93	0.95	0.95	0.74	0.81	0.90	0.94	0.89	0.8569	0.8890
Frontier Proprietary										
GPT-6 Astra	0.94	0.95	0.95	0.76	0.78	0.91	0.94	0.90	0.8589	0.8924
Claude Opus 5.5	0.95	0.96	0.96	0.81	0.81	0.92	0.95	0.92	0.8828	0.9097
Gemini 4 Argon	0.96	0.95	0.95	0.86	0.86	0.93	0.95	0.93	0.9087	0.9232
Large Open-Weight										
DeepSeek-V4.1-Flash	0.95	0.96	0.95	0.79	0.79	0.92	0.94	0.91	0.8716	0.9011
GLM-5.3	0.96	0.96	0.96	0.85	0.83	0.93	0.95	0.93	0.9007	0.9213
MiniMax-M3	0.97	0.97	0.96	0.87	0.85	0.93	0.96	0.95	0.9128	0.9310
Kimi-K3	0.96	0.96	0.96	0.83	0.83	0.92	0.95	0.93	0.8953	0.9167
MiMo-V2.6-Pro	0.97	0.97	0.96	0.87	0.85	0.93	0.96	0.95	0.9157	0.9324
DeepSeek-V4-Pro	0.96	0.96	0.96	0.85	0.84	0.93	0.95	0.93	0.9022	0.9222
Qwen3.8-2.4T-A95B	0.98	0.98	0.97	0.90	0.88	0.94	0.96	0.95	0.9313	0.9448
Small Open-Weight (4 Families)										
Qwen-4B	**1.00**	1.00	0.98	0.93	0.91	0.96	0.97	0.96	0.9572	0.9624
Qwen-9B	1.00	**1.00**	0.99	0.93	0.91	0.96	0.97	0.96	0.9595	0.9651
Qwen-27B (int8)	1.00	1.00	**1.00**	0.93	0.92	0.95	0.97	0.96	0.9609	0.9649
Gemma-E4B	0.98	0.97	0.96	**1.00**	0.90	0.95	0.96	0.95	0.9604	0.9571
Gemma-12B	0.98	0.97	0.97	0.94	**1.00**	0.95	0.96	0.94	0.9734	0.9640
Llama-3.2-1B	0.98	0.98	0.98	0.81	0.91	**1.00**	0.99	0.97	0.9229	0.9541
Llama-3.1-8B	0.99	0.99	0.99	0.83	0.92	1.00	**1.00**	0.99	0.9311	0.9617
Mistral-7B-v0.3	0.99	0.99	0.98	0.85	0.91	0.98	0.99	**1.00**	0.9322	0.9596

Table 9: Cross-domain replication on HC3 Wikipedia CS/AI ($N = 50$ prompts) under realistic nucleus filtering ($p = 0.99, K = 100$).

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Source Generator	Qwen-4B	Qwen-9B	Gemma-E4B	Gemma-12B	Sampled Mean	Greedy Mean	Sampled − Greedy Δ
Human (HC3 ELI5, detok.)	0.91	0.91	0.72	0.81	0.8374	0.8374	—
Qwen-4B (sampled $T = 1.0, p = 0.95$)	**1.00**	0.99	0.90	0.86	0.9396	0.9524	−0.0129
Qwen-9B (sampled $T = 1.0, p = 0.95$)	1.00	**1.00**	0.91	0.87	0.9449	0.9538	−0.0089
Gemma-12B (sampled $T = 1.0, p = 0.95$)	0.98	0.98	0.93	**1.00**	0.9739	0.9764	−0.0026
Gemma-E4B (sampled $T = 1.0, p = 0.95$)	0.97	0.96	**1.00**	0.85	0.9470	0.9583	−0.0113
GPT-6 Astra	0.92	0.93	0.74	0.72	0.8255	0.8255	—
Claude Opus 5.5	0.95	0.95	0.81	0.78	0.8728	0.8728	—
Gemini 4 Argon	0.95	0.95	0.84	0.81	0.8897	0.8897	—

Table 10: Stochastic-regeneration decoding ablation on HC3 ELI5 (realistic $p = 0.99, K = 100$, detokenized human reference). Local open-weight sources are regenerated using ancestral nucleus sampling ($T = 1.0, p = 0.95$) matched to the API sampling regime of Frontier Proprietary and Large Open-Weight models.

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A Additional Filter-Regime Matrices

Tables 11 and 12 report the full per-source and per-scorer HC3 ELI5 matrices under each scorer's shipped generation defaults and under unscaled ($T = 1.0$) shipped top-k/top-p filters.

B Zipf-Adjusted Lexical Category and Loss-Rate Taxonomy

To understand *which* words fail the nucleus filter across generator tiers—and why raw counts of lost words can be misleading without frequency normalization—we conducted a word-level lexical audit across the expanded 4-family HC3 ELI5 evaluation suite (9 scorers evaluating 20 sources at realistic $p = 0.99$, excluding self-scoring diagonal pairs $s = g$, totaling 2,586,369 off-diagonal word evaluations and 187,895 lost word instances).

Methodology & Word-Class Partition. For every off-diagonal scorer–source pair (g, s) with $g \neq s$ and every prompt $q \in \{1, \ldots, 50\}$, we align the whitespace word groups $W(y_{q,s})$ with the token-level survival mask under realistic $p = 0.99$ ($K = 100$). Each word instance w is normalized by lowercasing and stripping leading/trailing ASCII punctuation (while preserving internal apostrophes and hyphens, and distinguishing ASCII straight apostrophes `'` from Unicode right single quotation marks `U+2019`) and assigned to one of three mutually exclusive macro-categories:

1. **Function / Stopwords:** Standard closed-class English determiners, prepositions, conjunctions, auxiliary verbs, and pronouns (85 canonical forms).

2. **Standalone Punctuation / Symbols:** Tokens consisting solely of non-alphanumeric characters bounded by whitespace (e.g., spaced em-dashes `--`, bullets, or markdown symbols `#`, `>`).

3. **Content / Lexical Words:** All remaining open-class nouns, lexical verbs, adjectives, adverbs, numbers, and hyphenated/contracted lexical forms.

For each generator tier and each normalized word type v, we record total off-diagonal evaluations $N(v)$, total lost instances $L(v)$, and the **Zipf-adjusted per-occurrence loss rate** $r(v) = L(v)/N(v)$.

Finding 1: Macro-Category Breakdown. Table 13 demonstrates that across all four generator tiers, **content/lexical words account for** 71.7%**–**77.4% **of all lost words** and suffer a per-occurrence loss rate $2.1\times$ to $2.9\times$ higher than function/stopwords:

- In **Human** writing, 17.86% of all content word tokens fail the $p = 0.99$ filter ($8{,}992/50{,}355$), compared to only 6.25% of function words ($2{,}461/39{,}384$).

- Across **Frontier Proprietary** (12.99% content vs. 5.88% function), **Large Open-Weight** (10.66% content vs. 4.94% function), and **Small Open-Weight** (5.64% content vs. 2.72% function), the hierarchy of lexical reachability is driven primarily by a progressive $3.2\times$ contraction in content-word loss rates ($17.86\% \rightarrow 12.99\% \rightarrow 10.66\% \rightarrow 5.64\%$).

Finding 2: Raw-Count Leaders vs. Zipf-Adjusted Rate Leaders. Table 14 contrasts the top lost words ranked by raw count against those ranked by per-occurrence loss rate (≥ 45 evaluations). Because function words like `the`, `a`, and `and` occur 20,000–56,000 times per tier, even a 1.5%–6.0% failure rate produces hundreds to thousands of raw losses. Normalizing by occurrence frequency reveals the true high-risk lexical items: Reddit register markers (`tl;dr` 100%, `edit` 77.8%) and hedging words (`depending` 57.8%, `whether` 48.9%, `generally` 46.7%) in Human answers; nuanced verbs and qualifiers (`acts` 68.9%, `ordinary` 68.7%, `roughly` 56.5%) in Frontier Proprietary answers; markdown headers (`#` 100%), conversational preambles (`eli5` 75.6%, `honestly` 59.7%, `great` 50.9%), and spaced em-dashes (`--` 45.0%) in Large Open-Weight answers; and family-specific openers/orthography (`alright` 88.1%, curly-apostrophe `let's` 56.3% and `it's` 31.3%) in Small Open-Weight answers.

Finding 3: Why Function Words Still Fail Occasionally. Inspecting the $6{,}446$ total lost occurrences of `the` and `a` across the 9-scorer matrix reveals three concrete mechanisms: (i) **contextual phrase/idiom lock-in** (39.0%, $2{,}513/6{,}446$), where the scorer assigns ≥ 0.95 probability to a single fixed continuation token (e.g., an idiom or multi-word expression), pushing any alternative determiner past the 0.99 boundary; (ii) **sentence-initial**

Source Generator	Qwen-4B	Qwen-9B	Gemma-E4B	Gemma-12B	Qwen-27B (int8)	4-bf16 Scorer Mean	Human − Source Δ (95% CI)
Human Reference							
Human (HC3 ELI5, detok.)	0.66	0.67	0.60	0.70	0.68	0.6591	—
Frontier Proprietary (API, $T = 1.0, p = 0.95$)							
GPT-6 Astra	0.66	0.69	0.62	0.60	0.70	0.6440	$+0.0151$ ($[-0.0011, +0.0309]$)
Claude Opus 5.5	0.72	0.75	0.71	0.68	0.77	0.7181	-0.0590 ($[-0.0731, -0.0453]$)
Gemini 4 Argon	0.74	0.75	0.75	0.73	0.77	0.7414	-0.0823 ($[-0.0973, -0.0679]$)
Large Open-Weight (API, $T = 1.0, p = 0.95$)							
DeepSeek-V4.1-Flash	0.72	0.75	0.69	0.65	0.76	0.7020	-0.0429 ($[-0.0591, -0.0271]$)
GLM-5.3	0.73	0.75	0.71	0.68	0.77	0.7189	-0.0598 ($[-0.0741, -0.0458]$)
MiniMax-M3	0.74	0.76	0.72	0.68	0.77	0.7259	-0.0668 ($[-0.0818, -0.0518]$)
Kimi-K3	0.73	0.76	0.73	0.70	0.79	0.7305	-0.0715 ($[-0.0875, -0.0546]$)
MiMo-V2.6-Pro	0.75	0.77	0.75	0.71	0.78	0.7458	-0.0867 ($[-0.1010, -0.0722]$)
DeepSeek-V4-Pro	0.76	0.79	0.75	0.71	0.80	0.7515	-0.0924 ($[-0.1090, -0.0761]$)
Qwen3.8-2.4T-A95B	0.78	0.81	0.77	0.74	0.82	0.7770	-0.1180 ($[-0.1327, -0.1039]$)
Small Open-Weight (Local Greedy)							
Qwen-4B	**1.00**	0.93	0.85	0.80	0.87	0.8923	-0.2332 ($[-0.2479, -0.2191]$)
Qwen-9B	0.93	**1.00**	0.85	0.80	0.90	0.8969	-0.2378 ($[-0.2528, -0.2231]$)
Qwen-27B (int8)	0.91	0.93	0.86	0.83	**1.00**	0.8827	-0.2236 ($[-0.2385, -0.2091]$)
Gemma-12B	0.84	0.84	0.86	**1.00**	0.85	0.8861	-0.2270 ($[-0.2417, -0.2124]$)
Gemma-E4B	0.83	0.81	**1.00**	0.79	0.80	0.8578	-0.1987 ($[-0.2137, -0.1842]$)

Table 11: HC3 ELI5 cross-model WCS matrix under each scorer's shipped generation defaults (Qwen 3.5: $T = 0.6, k = 20, p = 0.95$; Gemma 4: $T = 1.0, k = 64, p = 0.95$).

Source Generator	Qwen-4B	Qwen-9B	Gemma-E4B	Gemma-12B	Qwen-27B (int8)	4-bf16 Scorer Mean	Human − Source Δ (95% CI)
Human Reference							
Human (HC3 ELI5, detok.)	0.77	0.79	0.60	0.70	0.80	0.7157	—
Frontier Proprietary (API, $T = 1.0, p = 0.95$)							
GPT-6 Astra	0.79	0.81	0.62	0.60	0.82	0.7084	$+0.0073$ ($[-0.0070, +0.0218]$)
Claude Opus 5.5	0.84	0.87	0.71	0.68	0.87	0.7758	-0.0601 ($[-0.0731, -0.0476]$)
Gemini 4 Argon	0.86	0.86	0.75	0.73	0.87	0.7983	-0.0826 ($[-0.0966, -0.0689]$)
Large Open-Weight (API, $T = 1.0, p = 0.95$)							
DeepSeek-V4.1-Flash	0.85	0.87	0.69	0.65	0.87	0.7618	-0.0461 ($[-0.0607, -0.0317]$)
GLM-5.3	0.85	0.87	0.71	0.68	0.88	0.7787	-0.0630 ($[-0.0764, -0.0496]$)
MiniMax-M3	0.85	0.87	0.72	0.68	0.87	0.7834	-0.0677 ($[-0.0811, -0.0541]$)
Kimi-K3	0.86	0.88	0.73	0.70	0.88	0.7897	-0.0740 ($[-0.0884, -0.0587]$)
MiMo-V2.6-Pro	0.87	0.89	0.75	0.71	0.89	0.8042	-0.0885 ($[-0.1016, -0.0753]$)
DeepSeek-V4-Pro	0.88	0.89	0.75	0.71	0.90	0.8070	-0.0913 ($[-0.1065, -0.0762]$)
Qwen3.8-2.4T-A95B	0.89	0.91	0.77	0.74	0.91	0.8292	-0.1135 ($[-0.1269, -0.1003]$)
Small Open-Weight (Local Greedy)							
Qwen-4B	**1.00**	0.98	0.85	0.80	0.95	0.9061	-0.1904 ($[-0.2038, -0.1774]$)
Qwen-9B	0.98	**1.00**	0.85	0.80	0.97	0.9092	-0.1935 ($[-0.2071, -0.1797]$)
Qwen-27B (int8)	0.97	0.98	0.86	0.83	**1.00**	0.9119	-0.1962 ($[-0.2096, -0.1829]$)
Gemma-12B	0.93	0.93	0.86	**1.00**	0.93	0.9311	-0.2154 ($[-0.2290, -0.2021]$)
Gemma-E4B	0.92	0.91	**1.00**	0.79	0.90	0.9065	-0.1908 ($[-0.2043, -0.1774]$)

Table 12: HC3 ELI5 cross-model WCS matrix under unscaled ($T = 1.0$) shipped top-k and top-p filters (Qwen 3.5: $T = 1.0, k = 20, p = 0.95$; Gemma 4: $T = 1.0, k = 64, p = 0.95$).

discourse framing (15.8%, $1{,}016/6{,}446$), where capitalized `The` or `A` competes against preferred bullet/discourse starters; and (iii) **mid-sentence determiner competition** (45.3%, $2{,}917/6{,}446$), where a scorer strongly prefers a bare plural or possessive pronoun over an article.

Generator Tier	Word Macro-Category	Lost Words	Total Evaluated	Category Loss Rate	Share of Tier's Lost Words
Human (HC3, 1 source)	Content / Lexical Words	8,992	50,355	17.86%	77.4%
	Function / Stopwords	2,461	39,384	6.25%	21.2%
	Standalone Punct. / Symbols	167	630	26.51%	1.4%
	Tier Total	**11,620**	**90,369**	**12.86%**	**100.0%**
Frontier Proprietary (3 sources)	Content / Lexical Words	34,897	268,587	12.99%	77.4%
	Function / Stopwords	9,959	169,281	5.88%	22.1%
	Standalone Punct. / Symbols	210	3,078	6.82%	0.5%
	Tier Total	**45,066**	**440,946**	**10.22%**	**100.0%**
Large Open-Weight (7 sources)	Content / Lexical Words	61,714	579,573	10.65%	73.5%
	Function / Stopwords	18,273	370,269	4.94%	21.8%
	Standalone Punct. / Symbols	3,941	14,427	27.32%	4.7%
	Tier Total	**83,928**	**964,269**	**8.70%**	**100.0%**
Small Open-Weight (9 sources, $s \neq g$)	Content / Lexical Words	34,062	604,144	5.64%	71.7%
	Function / Stopwords	13,242	487,664	2.72%	27.9%
	Standalone Punct. / Symbols	198	2,136	9.27%	0.4%
	Tier Total	**47,502**	**1,093,944**	**4.34%**	**100.0%**

Table 13: Macro-category breakdown of word losses at realistic $p = 0.99$ across the 4-family HC3 ELI5 suite (9 scorers, 20 sources, off-diagonal pairs $s \neq g$, 2,586,369 word evaluations).

Generator Tier	Top Lost Words by Raw Count (Lost / Total, Loss Rate)	Top Lost Words by Loss Rate (≥ 45 Evals: Rate, Lost / Total)
Human (HC3)	`the` (139/4,725, 2.9%), `a` (109/2,520, 4.3%), `and` (106/2,025, 5.2%), `in` (101/1,458, 6.9%), `that` (74/1,431, 5.2%), `of` (72/2,466, 2.9%), `i` (72/333, 21.6%), `this` (68/648, 10.5%), `from` (63/504, 12.5%)	`tl;dr` (100.0%, 45/45), `edit` (77.8%, 42/54), `depending` (57.8%, 26/45), `whether` (48.9%, 22/45), `set` (48.9%, 22/45), `volume` (48.9%, 22/45), `due` (48.9%, 22/45), `everybody` (48.9%, 22/45), `generally` (46.7%, 21/45)
Frontier Prop.	`the` (810/20,340, 4.0%), `a` (720/15,174, 4.7%), `and` (655/10,890, 6.0%), `that` (411/4,833, 8.5%), `your` (376/4,734, 7.9%), `it` (306/7,002, 4.4%), `in` (292/4,194, 7.0%), `you` (291/6,462, 4.5%), `is` (282/6,588, 4.3%)	`acts` (68.9%, 31/45), `ordinary` (68.7%, 68/99), `surprise` (57.4%, 31/54), `roughly` (56.5%, 61/108), `here's` (54.0%, 68/126), `recognize` (53.3%, 24/45), `neither` (51.9%, 28/54), `electronic` (51.1%, 23/45), `plenty` (51.1%, 23/45)
Large Open-Wt.	`--` (3,190/7,092, 45.0%), `the` (1,804/43,128, 4.2%), `a` (1,276/31,005, 4.1%), `and` (1,121/24,048, 4.7%), `your` (711/9,936, 7.2%), `so` (710/8,154, 8.7%), `that` (704/10,872, 6.5%), `here's` (668/1,521, 43.9%), `it` (572/16,101, 3.6%)	`#` (100.0%, 99/99), `eli5` (75.6%, 34/45), `haitian` (62.2%, 28/45), `honestly` (59.7%, 43/72), `deliberately` (55.6%, 45/81), `whose` (54.2%, 39/72), `hanging` (53.3%, 24/45), `borrowed` (51.9%, 28/54), `great` (50.9%, 357/702)
Small Open-Wt.	`the` (997/56,080, 1.8%), `and` (710/25,384, 2.8%), `is` (640/21,216, 3.0%), `a` (591/40,272, 1.5%), `you` (484/22,416, 2.2%), `it` (480/20,624, 2.3%), `to` (444/30,904, 1.4%), `so` (442/6,528, 6.8%), `imagine` (375/3,680, 10.2%)	`alright` (88.1%, 155/176), `let's` (curly, 56.3%, 27/48), `removes` (39.6%, 19/48), `poland` (35.4%, 17/48), `-` (32.3%, 31/96), `it's` (curly, 31.3%, 223/712), `asked` (29.2%, 35/120), `immediately` (28.6%, 32/112), `grown-ups` (26.6%, 34/128)

Table 14: Comparison of top lost words ranked by raw failure count (left) versus Zipf-adjusted per-occurrence failure rate among words with ≥ 45 evaluations (right) at realistic $p = 0.99$ across the 9-scorer, 20-source 4-family ELI5 pool.